

NLP FINAL PROJECT

Fine Grained Narrative Classification

Problem Statement

As digital news media has drastically increased people's ability to get news, it has also extended the spread of biased and propagandistic journalism with articles framed to influence public attitudes toward an issue. Traditional analysis focuses on relatively shallow aspects of an article such as surface sentiment and/or isolated techniques but often do not take into account the deeper narrative structures that shape the presentation of information.

A narrative is best described as being the underlying storyline that ties together the facts and opinions of an article into a coherent ideological message, thereby legitimizing one's own view while delegitimizing the view on the other side of an issue. Consequently, using narrative structure as a basis for determining media bias will require the identification of fine-grained narratives rather than simply using a basic classification scheme.

The problem addressed in this project is fine-grained narrative classification, which uses a dataset annotated with event types, bias, and narrative labels. The problem can be defined as a hierarchical classification problem, which consists of the following tasks:

- **Article Bias Classification:** Articles are classified as Pro-Government, Pro-Opposition, or Neutral.
- **Narrative Classification:** Articles classified as biased are then classified according to the corresponding event-type narrative label.

This is a challenging task due to the implicit and contextual nature of narratives, and because narrative prediction is dependent on bias classification, where "neutral" articles have no narratives. The goal is to develop baseline models for this task and assess its efficacy, leading to a greater comprehension of how news is being framed.

Dataset Description

This dataset for this project is a modification of the INDI-PROP dataset that was specifically developed to detect propaganda and bias in Indian news media. The data consists of annotated articles from prominent socio-political events in India, namely the Citizenship Amendment Act (CAA) and the Farmers' Protest. The rationale for using these two events is that they are highly polarised on many levels and elicit vast amounts of media coverage and a significant amount of differing ideologies.

Dataset Structure

In the current dataset, there are multiple examples of articles and each has certain characteristics:

- Article text : The complete text of each article.
- Event : The event associated with the article (CAA/Farmers Protests).
- Bias label : The bias label for each article represents the ideological position of the article: Pro government , Pro opposition or Neutral
- Narrative label :The narrative label provides an indication of the finer detail of the narrative of the article. It is only assigned to biased articles as neutral articles will only have the label of 'No Narrative'.
- Persuasive techniques : Labels for various propaganda techniques (for instance : loaded language, appeal to emotion, name calling).

Annotation Scheme

The dataset employs a hierarchical annotation framework with three levels:

1. Bias Annotation: Captures the overall ideological stance of the article based on tone, framing, and attribution.
2. Narrative Annotation: Identifies the specific narrative constructed within the article (e.g., glorification or vilification).
3. Persuasive Technique Annotation: Detects the rhetorical techniques used to reinforce the narrative or bias.

This hierarchical structure allows for analyzing how bias, narratives, and persuasive techniques interact within news discourse.

Methodology

1. Baselines (Mid project submission) :

To establish a strong reference for performance, we implement transformer-based baseline models using **BERT** and **RoBERTa**. These models were fine-tuned on the INDI-PROP dataset for the three classification tasks: bias classification , narrative classification , and persuasive technique identification.

Model Selection

We used the following pretrained models:

- **BERT (bert-base-uncased)**
- **RoBERTa (roberta-base)**

Both models are widely used in natural language processing tasks and provide strong contextual representations of text. A unified implementation pipeline is used for both models, where only the model name and task-specific configurations are modified, ensuring consistency across experiments.

Input Representation and Preprocessing

Each news article is tokenized using the corresponding pretrained tokenizer. For label processing:

- **Single-label classification (Bias):** Labels are encoded as integer values.
- **Multi-label classification (Narrative and Technique):** Labels are represented as binary vectors, where each position corresponds to the presence or absence of a specific class.

Fine-tuning Setup

The models are fine-tuned using the HuggingFace Transformers framework. The following hyperparameters are used consistently across all experiments:

- **Learning rate:** 2e-5
- **Batch size:** 8
- **Number of epochs:** 5
- **Weight decay:** 0.01

Task Formulation

The problem is formulated differently depending on the task:

- **Bias Classification:**
A single-label classification task with three classes (Pro-Government, Pro-Opposition, Neutral).
- **Narrative and Technique Classification:**
Multi-label classification tasks where each article may belong to multiple categories simultaneously. A sigmoid activation is applied to model outputs, and predictions are obtained using a threshold of **0.2**, determined empirically.

Evaluation Metrics

Different evaluation metrics are used based on the task:

- **Bias Classification:** Macro F1-score, precision, recall and accuracy
 - **Narrative and Technique Classification:** Macro F1-score, precision, recall and accuracy.
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2. Original Paper's Approach :

The original paper proposes a structured framework called **FANTA (Bias & Fine-Grained Narrative Classification)** for analyzing biased news articles. The approach is designed to capture the hierarchical nature of media bias by modeling the relationship between bias, narrative framing, and persuasive techniques.

FANTA follows a multi-step pipeline execution that resembles how a human would reason when interpreting news content. Instead of directly predicting fine-grained labels, the framework decomposes the task into several intermediate steps which include :

1. **Information Extraction:**
The model first extracts key entities and contextual information from the article using techniques such as Named Entity Recognition (NER), coreference resolution, and relation extraction.
2. **Context Framing:**
The extracted information is used to understand how the article frames events and entities, providing contextual grounding for subsequent classification tasks.
3. **Bias Classification:**
The article is categorized into one of three classes : Pro-Government, Pro-Opposition, or Neutral. This step captures the overall ideological stance of the article.
4. **Fine-grained Narrative Classification:**
Based on the identified bias and context, the model assigns one or more narrative labels from an event-specific taxonomy. These narratives represent the underlying ideological framing of the article.

5. **Persuasive Technique Classification:**

In addition to narrative classification, the paper also introduces a framework for identifying persuasive techniques. These techniques are categorized into a set of predefined classes and the model predicts their presence at the span or article level.

3. Our Proposed Method:

1. **Problem Setup**

We treat narrative detection as a multi-label classification task:

- Input: a news article
- Output: one or more narrative labels

We use a, single-stage model , to directly predict narratives.

2. **Model Overview**

We use a transformer encoder (DeBERTa) to convert each article into a vector.

We then use two classification heads:

(a) Description-based head

Each label has a text description (definition + examples)

Labels are encoded into vectors

Prediction = similarity between article and label vectors

Helps for rare labels

(b) Linear head

Standard classifier trained on data

Helps for frequent labels

(c) Fusion

Final score = weighted combination of both heads

Model learns how much to trust each head per label

3. Training

- Use binary cross-entropy with focal weighting
- Handles class imbalance and hard examples
- Ignore labels with no training data

4. Thresholding

- Use per-label thresholds instead of a fixed 0.5
- Tune each threshold on validation data
- Improves performance, especially for rare classes

5. Data Handling

- Stratified train/validation split
- Augment rare classes by dropping parts of text
- Keep neutral articles as negative examples

6. Training Setup

Lower learning rate for encoder, higher for classifier heads

Use AdamW, scheduling, gradient clipping, early stopping

7. LLM-Based Reranking (Post-Processing)

After classification, we apply an LLM-based reranking step to improve prediction quality.

(a) Candidate Generation

- Convert logits to probabilities
- Select candidate labels using a *looser threshold* (lower than the tuned threshold)
- Ensures *high recall*, especially for rare classes

(b) LLM Filtering

Provide the LLM with:

1. article text
2. candidate labels
3. label descriptions
4. model confidence scores

The LLM acts as a filter, not a generator:

- it can remove incorrect labels
- it cannot add new labels

Prompt encourages:

- keeping labels unless clearly incorrect
- avoiding over-aggressive filtering

(c) Safe Fusion

Final predictions combine:

1. original model predictions
2. LLM-filtered labels

High-confidence predictions from the model are always preserved

LLM only refines uncertain cases

8. Key Design Principle

The system is designed as:

Model → High Recall → LLM → Precision Refinement

This ensures:

- model captures as many relevant labels as possible
- LLM improves precision without harming recall

Results and Analysis

Model	Event	Micro	Macro	Weighted	Oracle Bias	Reranker
DeBERTa (paper)	CAA	0.538	0.269	0.471	no	None
Ours	CAA	0.546	0.513	0.653	no	Llama-3.3-70B
FANTA-GPT (4o)	CAA	0.670	0.660	0.700	yes	–
DeBERTa (paper)	Farmers	0.669	0.221	0.637	no	None
Ours	Farmers	0.260	0.221	0.339	no	Llama-3.3-70B
FANTA-GPT (4o)	Farmers	0.730	0.700	0.780	yes	–

On CAA, our system achieves macro F1 of 0.513, compared to 0.269 for the DeBERTa baseline reported in the original paper ; a gain of +0.244 macro F1 points. This improvement is attributable to our hierarchy-aware focal loss and LLM-based reranking, which together improve rare-class recall without sacrificing majority-class precision.

On the Farmers' protest, macro F1 matches the paper's DeBERTa baseline (0.221) despite operating under strictly harder conditions: no oracle bias labels, extreme class imbalance (three classes with fewer than 10 training examples), and a significantly weaker reranking model (Llama-3.3-70B via Groq vs. GPT-4o-mini). F4, F8, and F9 , which together account for only 17 training examples across three classes , score zero F1, consistent with what any model would produce under such sparsity. Excluding these three zero-resource classes, our per-class F1 averages 0.315, suggesting the architecture performs reasonably on classes with minimal training signal.